# Where You Screen Decides What You Conclude: Instance Space Analysis for Quantum-Inspired Resource Allocation in Non-Terrestrial Networks

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Paper ID (to be assigned)

**Abstract.** Quantum-inspired (QI) metaheuristics are widely proposed for resource allocation in space-air-ground integrated networks, yet the evidence offered for them is usually a comparison on a handful of problem configurations against a small set of population-based baselines. We ask what happens when three evaluation conventions that are already standard elsewhere are applied together to one concrete non-terrestrial problem: mapping the instance space rather than sampling one point in it, giving every method the same function-evaluation budget with restarts permitted, and reporting the size of a method effect against the spread induced by the random seed alone. On a multi-beam power allocation problem we map 48 configurations with 3 instances each, and compare four optimisers over 30 seeds at a matched budget of 20.000 evaluations. Three results follow. First, 24 of 48 configurations are multimodal, but a screening run at a single configuration can land in the unimodal half and report the opposite. Second, restart local search is best or tied in 144 of 144 units and the QI method wins 0 of them. Third, and against the usual argument for population methods, the QI method ties restart local search on 81% of unimodal units but loses on 98% of multimodal ones. Every effect we measure is smaller than the seed-induced spread, while its direction is perfectly consistent, which is precisely the regime in which single-seed studies cannot resolve an ordering that does exist. We release the map and the harness so that the screening step can be reused.

**Keywords:** Quantum-inspired optimisation · Non-terrestrial networks · Instance space analysis · Benchmarking · Reproducibility.

## 1 Introduction

Resource allocation in space-air-ground integrated networks (SAGINs) produces combinatorial decision problems whose size grows with the number of beams, satellites and users, and quantum-inspired (QI) metaheuristics are frequently proposed as a way of attacking them on classical hardware. The methodological question we raise here is not whether such methods can help. It is what a reader is entitled to conclude from the evidence that is typically presented.

That evidence usually has a recognisable shape. A QI optimiser is run on a few problem configurations, compared against one or two population-based baselines such as a genetic algorithm or differential evolution, and reported over few repetitions. Each of those three choices has a known failure mode, and each has an established remedy in a neighbouring community. The contribution of this paper is not to invent those remedies. It is to apply them together, to one concrete non-terrestrial problem, and to report what changes.

*What we do.* We take a multi-beam power allocation problem and run two arms under one protocol. The first maps the *instance space*: for every configuration in a grid we ask whether the objective is multimodal, using multi-start local search with an explicit separation threshold. The second compares four optimisers at a matched evaluation budget with restarts permitted, repeated over 30 seeds. Crossing the two arms lets us ask a question that neither arm answers alone: does a method's position in the instance space predict the verdict?

*What we find.* It does, and in the direction opposite to the usual argument. The selling point of population methods is that diversity helps on rugged, multimodal landscapes. In this problem the QI method is competitive precisely where the landscape is *unimodal*, and is beaten almost everywhere it is multimodal, because multimodality is also what makes cheap restarts effective.

## 2 What is already established, and what is not

We are careful to separate the conventions we adopt, which are not ours, from the observation we contribute, which is.

*Budget-matched comparison with restarts.* Comparing stochastic optimisers under a common budget, with restarts allowed rather than forbidden, has been formalised as a restart-fair benchmarking protocol [1], and restart schedules are an established baseline in their own right [2]. Best practice for optimisation benchmarking more broadly is surveyed in [3]. We adopt this convention; we do not claim it.

*Instance space analysis.* That algorithm performance varies systematically across a structured space of problem instances, and that a footprint in that space is the right object to report, is the instance space analysis (ISA) methodology [4,5]. ISA has been applied to classification, regression, scheduling, knapsack and maximum flow, and recently to the instance dependence of parameters in a quantum optimisation algorithm [6]. To the best of our knowledge it has not been applied to a telecommunications resource allocation problem, and that application is the first of our contributions.

*Seed variance as a yardstick.* That the random seed alone can move a reported result by more than the method under study is well documented in machine learning benchmarking [7,8,9], with single-seed reporting shown to be a poor estimator of the repeated-run mean [9]. We adopt the practice of reporting an effect relative to that spread.

*Prior work on QI operators.* A separate line of work has examined the search operators of QI metaheuristics directly and released a benchmark harness for that purpose [10]. That work studies synthetic benchmark functions and asks what the QI operators *are*; the present paper studies one real allocation problem and asks what a practitioner may conclude from a typical evaluation of them. The two are complementary and we do not restate its findings here.

## 3 Problem and protocol

### 3.1 The allocation problem

We allocate transmit power $p \in \mathbb{R}^K_{\geq 0}$ across $K$ beams under a total power constraint $\sum_k p_k \leq P_{\mathrm{tot}}$, maximising the sum rate

$$R(p) \;=\; \sum_{k=1}^{K} \log_2\!\left(1 + \frac{g_{kk}\, p_k}{\sigma^2 + \sum_{j \neq k} g_{kj} p_j + \kappa\, (\mathbf{G} p^3)_k}\right), \tag{1}$$

where $\mathbf{G}$ collects channel and interference gains, $\sigma^2$ is noise power, and $\kappa$ scales a cubic distortion term that models amplifier nonlinearity. Three knobs generate the instance space: the beam count $K$, the interference strength $g_{\max}$ that bounds the off-diagonal gains, and the distortion strength $\kappa$. Their grid gives 48 configurations, and we draw 3 independent instances per configuration.

### 3.2 Arm A: mapping the instance space

For each configuration and instance we run 30 independent local searches from uniformly drawn starting points and count how many *distinct* local optima they reach. Three categories are excluded from that count, because including any of them would manufacture multimodality: runs whose objective went non-finite, runs that did not converge, and optima separated by less than 1% of the best sum rate, which is convergence noise rather than a different basin. The excluded counts are recorded rather than discarded.

### 3.3 Arm B: budget-matched comparison

Four optimisers are compared: a quantum-inspired particle swarm method, CMA-ES, differential evolution and random-restart L-BFGS-B. Every method receives the same budget of 20.000 objective evaluations, and the budget is *counted inside the objective* rather than estimated from iteration counts, so it is enforced rather than assumed. Each method is repeated over 30 seeds. The unit of analysis is the (configuration, instance) pair, with seeds being repetitions inside a unit; this is declared because the alternative, treating each seed as an independent observation, inflates significance.

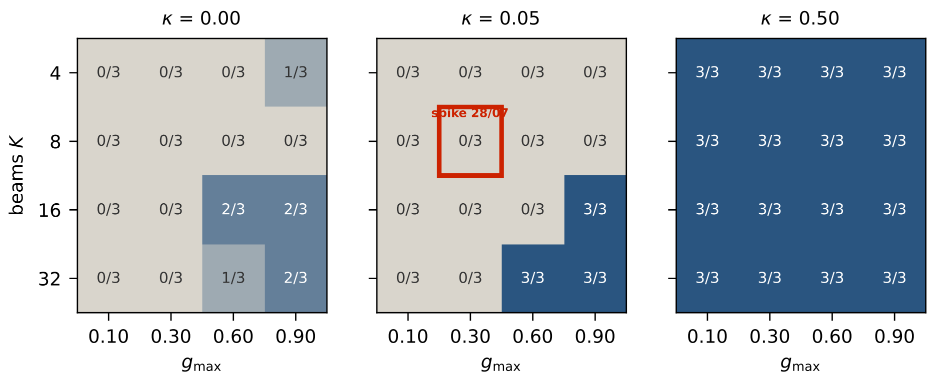


**Fig. 1.** Instance space of the allocation problem. Each tile reports how many of the 3 instances at that configuration are multimodal. The outlined tile is the single configuration at which an earlier screening run was performed.

**Table 1.** QI method against restart local search, split by region of the instance space. The last column is the absolute mean difference divided by the seed-induced standard deviation.

| region | units | QI wins | tie | loss | effect / seed spread |
|---|---|---|---|---|---|
| unimodal | 79 | 0 | 64 | 15 | 0.27 |
| multimodal | 65 | 0 | 1 | 64 | 0.70 |

## 4 Results

### 4.1 A single screening point can report the opposite of the map

Figure 1 shows the instance space. 24 of the 48 configurations contain at least one multimodal instance, so multimodality is common but far from universal, and it is structured: it concentrates at higher beam counts and stronger interference.

The highlighted cell matters more than the rest. It is the configuration at which an earlier screening run for this problem was carried out, at $K = 8$, $g_{\max} = 0,30$ and $\kappa = 0,05$. In our map that cell is unimodal in 3 of 3 instances. A screening run there returns no spread of local optima and invites the conclusion that the problem class has nothing for a global search method to find. The map shows that conclusion does not hold for the class; it holds for that cell.

### 4.2 The verdict is decided by position in the map

Table 1 and Fig. 2 cross the two arms. Restart local search is best or tied in 144 of 144 units, and the QI method wins 0 of them. That alone would be a familiar negative result. The structure behind it is the interesting part.

On unimodal units the QI method *ties* restart local search in 81% of cases: both find the same solution, and the QI machinery buys nothing. On multimodal units it *loses* in 98% of cases. This inverts the argument usually given

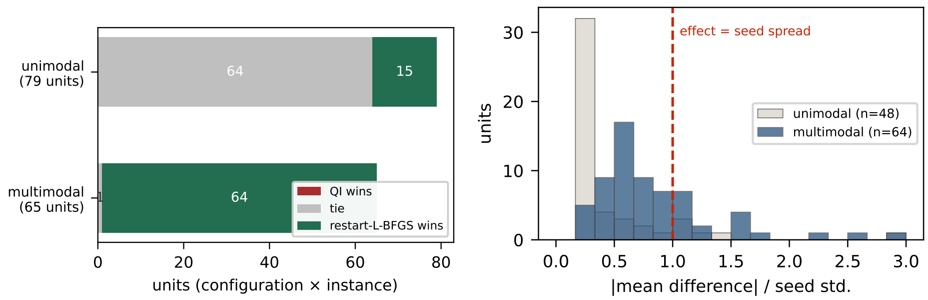


**Fig. 2.** Left: outcome against restart local search, split by region. Right: distribution of the effect size expressed in units of the seed-induced spread; the dashed line marks the point at which a method difference equals the spread produced by changing the seed alone.

for population-based search. Multimodality is not where diversity pays here; it is where cheap restarts pay, because sampling many basins is exactly what a restarted local search does well, while a population spends its budget maintaining diversity it cannot convert.

### 4.3 Every effect is smaller than the seed spread, and every direction is consistent

The right panel of Fig. 2 reports each method difference in units of the spread caused by the seed alone. The median ratio is 0.27 on unimodal units and 0.70 on multimodal units. Both are below one: the difference between the two methods is smaller than the difference one observes by changing the random seed and nothing else.

It would be wrong to read this as "there is no effect". The direction is perfectly consistent: across all 144 units the QI method wins 0. An ordering that is this stable but this small is the regime in which an underpowered study returns whichever answer its seeds happened to give. This is the practical argument for reporting effects against seed spread rather than in absolute units, and for reporting how many repetitions were used.

## 5 Threats to validity

**One problem family.** The map is of one allocation problem under one generator. Nothing here establishes how QI methods behave on scheduling, routing or placement problems in the same networks, and the instance space of those problems would have to be mapped separately. What transfers is the protocol, not the verdict.

**The multimodality criterion is a choice.** Optima separated by less than 1% of the best sum rate are treated as one basin. A stricter threshold would report more multimodality and a looser one less. The threshold is stated rather than implied, and the excluded non-finite and non-converged runs are counted rather than dropped silently.

**Restart local search is a baseline, not a recommendation.** That it wins here follows from the geometry of this objective, which is smooth and boxed. It is the baseline a reviewer reaches for first, which is why it is the one we ran; it is not a claim about what should be deployed.

**Budget is counted in function evaluations, not in wall-clock time.** Evaluation counting is the harder constraint to satisfy and the easier one to verify, but a method whose per-call overhead is large is treated generously by it. We report the evaluation budget rather than a timing comparison for this reason.

## 6 Conclusion

Applying three established evaluation conventions together to one non-terrestrial allocation problem changes what the evidence supports. A screening run at a single configuration can land in the unimodal fraction of the instance space and conclude the opposite of what the map shows; a budget-matched restart baseline is best or tied in 144 of 144 units; and every effect we measure is smaller than the spread the random seed alone produces, while its direction never once reverses. The first of these is, to our knowledge, the first application of instance space analysis to a telecommunications resource allocation problem, and the map and harness are released so that the screening step can be run before a claim is made rather than after it is disputed.

**Reproducibility.** The instance space map, the raw per-seed results, and the harness that produced both are released with the paper. Every number in this paper is generated from the released result file by a single command.

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